Ejercicio 2: $h[n] = \delta[n] - \delta[n-4]$. Hallar $y[n] = x[n] * h[n]$

a) $x[n] = \cos(\omega_0 n T_s)$

RTA: $A\cos(\omega_0 n T_s + \varphi)$

PROP: $\left\{\begin{array}{l} x[n] * \delta[n] = x[n] \\ x[n] * \delta[n-k] = x[n-k] \end{array}\right\}$ $y[n] = x[n] * \delta[n] - x[n] * \delta[n-4]$

$\Rightarrow$ $y[n] = x[n] - x[n-4]$ Para los 3 ítems

$x[n] = \cos(\omega_0 n T_s)$

$y[n] = \cos(\omega_0 n T_s) - \cos(\omega_0 (n-4) T_s)$

$y[n] = \cos(\omega_0 n T_s) - \cos(\omega_0 n T_s - 4\omega_0 T_s)$

Identidad Trigonométrica

$\cos A - \cos B = -2\,\text{sen}\left(\frac{A+B}{2}\right)\text{sen}\left(\frac{A-B}{2}\right)$

$A = \omega_0 n T_s$ ; $B = \omega_0 n T_s - 4\omega_0 T_s$

$y[n] = -2\,\text{sen}(\omega_0 n T_s - 2\omega_0 T_s)\,\text{sen}(2\omega_0 T_s)$ → Lo quiero en cosenos

$\Rightarrow$ uso que $\text{sen}(x) = \cos(x - \pi/2)$

$y[n] = \underbrace{-2\,\text{sen}(2\omega_0 T_s)}_{A}\cos(\omega_0 n T_s \underbrace{- 2\omega_0 T_s - \pi/2}_{\varphi})$

$\boxed{y[n] = A\cos(\omega_0 n T_s - \varphi)}$ con $A = -2\,\text{sen}(2\omega_0 T_s)$ y $\varphi = -(2\omega_0 T_s + \pi/2)$

b) $x[n] = \left(\frac{1}{2}\right)^n u[n]$

$\boxed{y[n] = (1/2)^n u[n] - (1/2)^{n-4} u[n-4]}$

$$y[n] = \begin{cases} 0 & n < 0 \\ (1/2)^n & 0 \leq n < 4 \\ (1/2)^n - (1/2)^{n-4} & n \geq 4 \end{cases}$$

c) $x[n] = u[n+1] - u[n-2]$

$x[n-4] = u[n-3] - u[n-6]$

$y[n] = (u[n+1] - u[n-2]) - (u[n-3] - u[n-6])$

$$\boxed{y[n] = u[n+1] - u[n-2] - u[n-3] + u[n-6]}$$